

TP2 Gestión de contenedores en DOCKER

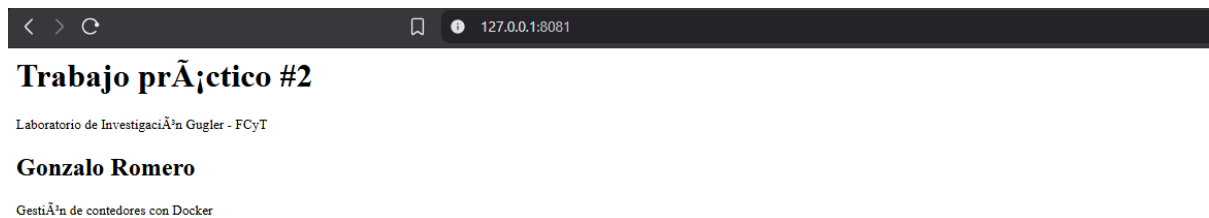
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Ejercicio 1: Dockerfile y su estructura

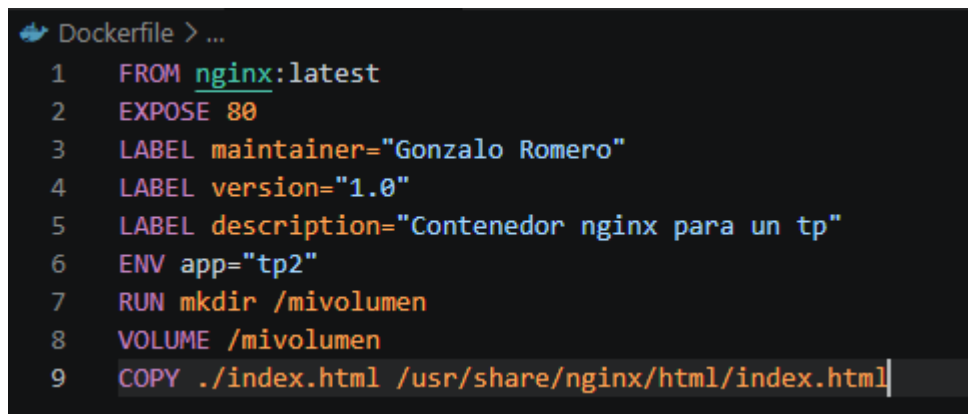
Crear una imagen Docker que ejecute una aplicación web que muestre un mensaje al acceder vía el puerto 8081 de su navegador web (use `http://127.0.0.1:8081`). Genere un archivo HTML como hicimos en el TP#1.

Genere un Dockerfile para esta imagen. Utilice las instrucciones: ENV , EXPOSE , LABEL , USER y VOLUME:

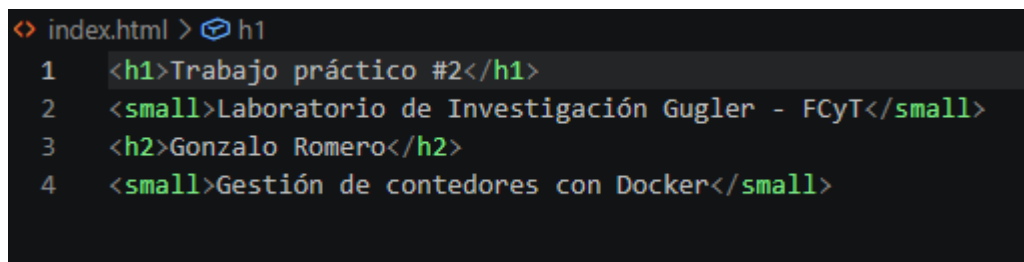
html abierto en 8081:



dockerfile:



html:



Ejercicio 2: Comandos básicos de Docker para la construcción

Muestre cómo utilizar los comandos pull y build para la construcción de la imagen para el Dockerfile anterior.

docker pull:

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2>docker pull nginx:latest
latest: Pulling from library/nginx
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

docker build:

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2>docker build -t tp2 .
[+] Building 1.6s (8/8) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 282B                               0.0s
=> [internal] load metadata for docker.io/library/nginx:latest  0.7s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                     0.0s
=> [1/3] FROM docker.io/library/nginx:latest@sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f56223 0.1s
=> => resolve docker.io/library/nginx:latest@sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f56223 0.1s
=> [internal] load build context                                0.1s
=> => transferring context: 32B                                    0.0s
=> CACHED [2/3] RUN mkdir /mivolumen                            0.0s
=> CACHED [3/3] COPY ./index.html /usr/share/nginx/html/index.html 0.0s
=> exporting to image                                           0.4s
=> => exporting layers                                           0.0s
=> => exporting manifest sha256:4a3dd34dd3e0f8c67f1a484aeca25cb5c3b5b4d73e4eda5e24f300f5e1d490 0.0s
=> => exporting config sha256:69d8ca2e970f69f1c5ee2aed584905aef55777d616cab85456caaa8dece24cdf 0.0s
=> => exporting attestation manifest sha256:de920c9cbce134faabf015951ff066cbc1bf388c370d95f7967267803906952f 0.1s
=> => exporting manifest list sha256:df9e81d9b8863c99b64f8fed3e690bc9e33ebc5917374282990ca0fe7bb1840d 0.1s
=> => naming to docker.io/library/tp2:latest                    0.0s
=> => unpacking to docker.io/library/tp2:latest                  0.1s
```

Ejercicio 3: Buenas prácticas para la creación de imágenes

Teniendo el siguiente Dockerfile optimicelo con buenas prácticas.

```
FROM nginx
COPY ./index.html /app
RUN apt update
RUN apt -y install vim
RUN apt -y install bash
LABEL version=1.0.0
WORKDIR /app
EXPOSE 80
```

```
FROM nginx
RUN apt update
RUN apt -y install vim
RUN apt -y install bash
LABEL version=1.0.0
WORKDIR /app
EXPOSE 80
COPY ./index.html /app
```

Ejercicio 4: Volúmenes y persistencia de datos

Crear un volumen y configure un contenedor nginx:latest para que la utilice.

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2>docker volume create volumen-tp2
volumen-tp2
```

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2>docker run -d -v volumen-tp2 --name contenedor-tp2 tp2
aa87be29d0afb41c56766e11f0eba2cbb9eb5319c0804b892056f883ad5b2549

C:\Users\Gonza\Desktop\tps\DOCKER\tp2>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
aa87be29d0af   tp2       "/docker-entrypoint..." 5 seconds ago  Up 4 seconds  80/tcp       contenedor-tp2

C:\Users\Gonza\Desktop\tps\DOCKER\tp2>
```

Ejercicio 5: Tipos de redes en Docker

Crear una red personalizada bridge y configure un contenedor nginx:latest para que la utilice

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2>docker network create red_tp2
b8e3e4b92173a3b4aea7ca1873efbc3994e3dfa409220d66f65cb40ffb082079
```

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2>docker network connect red_tp2 contenedor-tp2
```

Ejercicio 6: Comunicación entre contenedores

Genere un contenedor python:3.12 y conéctelo con la red del ejercicio anterior.

dockerfile:

```
py > Dockerfile > ...
1 FROM python:3.12
2 WORKDIR /app
3 COPY requirements.txt .
4 RUN pip install -r requirements.txt
5 COPY . .
6 EXPOSE 5000
7 CMD ["python", "pruebatp2.py"]
```

archivo .py:

```
> pruebatp2.py > ...
1 from flask import Flask
2 app = Flask(__name__)
3 @app.route('/')
4 def hello():
5     return "Hola, Docker"
6 if __name__ == "__main__":
7     app.run(host='0.0.0.0', port=5000)
```

requirements.txt:

```
py > requirements.txt
1 flask
```

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2\py>docker build -t python_tp2 .
[+] Building 8.1s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 186B                               0.0s
=> [internal] load metadata for docker.io/library/python:3.12     1.1s
=> [auth] library/python:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> CACHED [1/5] FROM docker.io/library/python:3.12@sha256:bd55e06128e2743d526738dd5cf5db924e49ead66d4fe2ad15f2aa 0.1s
=> => resolve docker.io/library/python:3.12@sha256:bd55e06128e2743d526738dd5cf5db924e49ead66d4fe2ad15f2aac8ad86a 0.1s
=> [internal] load build context                                  0.1s
=> => transferring context: 445B                                   0.0s
=> [2/5] WORKDIR /app                                             0.1s
=> [3/5] COPY requirements.txt .                                  0.2s
=> [4/5] RUN pip install -r requirements.txt                      3.9s
=> [5/5] COPY . .                                                 0.2s
=> exporting to image                                             2.0s
=> => exporting layers                                           1.0s
=> => exporting manifest sha256:62b7ea0025547091cabbf2b31c6e1411f4f2d8927c5c5a1fbaf432c210175159 0.1s
=> => exporting config sha256:00f1365fe5e8b32fcff404dca8b91f9a5e0afa05d0773ee592e9e7b387eee8cf 0.1s
=> => exporting attestation manifest sha256:5794ad3cb80305cf02d903f507358b59458f0b2b381fc26bafaac249a0eaf48e 0.1s
=> => exporting manifest list sha256:3b1101ee0eaab01903c2ec6a2445e4d0cbb37fe61cde55a1eee2e1dee0100cf 0.1s
=> => naming to docker.io/library/python_tp2:latest              0.0s
=> => unpacking to docker.io/library/python_tp2:latest           0.6s
```

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2\py>docker run -d -p 5001:5000 --name contenedor_tp2_python python_tp2
717a863c495b844393db5ea67c4114cab862f4b1dbfae35a5e0a293aa5353a40
```

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2\py>docker network connect red_tp2 contenedor_tp2_python
```

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2\py>docker network inspect red_tp2
[
  {
    "Name": "red_tp2",
    "Id": "b8e3e4b92173a3b4aea7ca1873efbc3994e3dfa09220d66f65cb40fffb082079",
    "Created": "2026-05-30T13:57:33.507208654Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {},
    "Containers": {
      "717a863c495b844393db5ea67c4114cab862f4b1dbfae35a5e0a293aa5353a40": {
        "Name": "contenedor_tp2_python",
        "EndpointID": "fd4359621f0ebbcbee080022b7982e744bb842b7f28568ca9eccf0ea02e20d7c",
        "MacAddress": "b2:d2:b2:25:bc:fd",
        "IPv4Address": "172.18.0.3/16",
        "IPv6Address": ""
      },
      "aa87be29d0afb41c56766e11f0eba2cbb9eb5319c0804b892056f883ad5b2549": {
        "Name": "contenedor-tp2",
        "EndpointID": "56a8d5e45704d36062edc34e7af70490d794ab221324ca4fd4fa425b11307313",
        "MacAddress": "ea:38:2f:59:99:a5",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      }
    },
    "Status": {
      "IPAM": {

```

Ejercicio 7: Docker Compose

Genere un archivo docker-compose.yml para comunicar dos servicios web que estén en la misma red (bridge).

archivo docker-compose.yml:

```
version: "3"

Run All Services
services:
  Run Service
  web:
    build: ./html
    ports:
      - "8081:80"
  Run Service
  app:
    build: ./py
    ports:
      - "5002:5000"

networks:
  red_tp2:
```

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2>docker compose up -d --build
time="2024-06-30T11:37:49-03:00" level=warning msg="C:\\Users\\Gonza\\Desktop\\tps\\DOCKER\\tp2\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 971B done
#1 DONE 0.0s

#2 [app internal] load build definition from Dockerfile
#2 transferring dockerfile: 186B done
#2 DONE 0.1s

#3 [web internal] load build definition from Dockerfile
#3 transferring dockerfile: 282B done
#3 DONE 0.1s

#4 [web internal] load metadata for docker.io/library/nginx:latest
#4 DONE 0.1s

#5 [app internal] load metadata for docker.io/library/python:3.12
#5 ...

#6 [web internal] load dockerignore
#6 transferring context: 2B done
#6 DONE 0.1s

#7 [web 1/3] FROM docker.io/library/nginx:latest@sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
#7 resolve docker.io/library/nginx:latest@sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6 0.1s done
#7 DONE 0.1s

#8 [web internal] load build context
#8 transferring context: 203B done
#8 DONE 0.2s

#9 [web 2/3] RUN mkdir /mivolumen
#9 CACHED

#10 [web 3/3] COPY ./index.html /usr/share/nginx/html/index.html
#10 CACHED

#11 [web] exporting to image
#11 exporting layers done
#11 exporting manifest sha256:9f2769d6882a6a0ab129448fd8e2a40be1054254f77481e66ee8c2cb40aa4bb42 0.1s done
#11 exporting config sha256:3123553ad877a1344e83b4ff410b3ae5af61a22efe068324c1ea260fe20740b3
#11 ...

#12 [auth] library/python:pull token for registry-1.docker.io
#12 DONE 0.0s
```

```

#11 [web] exporting to image
#11 exporting config sha256:3123553ad877a1344e83b4ff410b3ae5af61a22efe668324c1ea260fe20740b3 0.1s done
#11 exporting attestation manifest sha256:6f3d6c7b3992ced8153cc3bb719c4d42a655dac089ae1c8549daa4efdd38b9
#11 exporting attestation manifest sha256:6f3d6c7b3992ced8153cc3bb719c4d42a655dac089ae1c8549daa4efdd38b9 0.1s done
#11 exporting manifest list sha256:52e98865f1074617c3e846d6d3afa6df106d73e6e1975a19bd996d7ef658574c
#11 exporting manifest list sha256:52e98865f1074617c3e846d6d3afa6df106d73e6e1975a19bd996d7ef658574c 0.1s done
#11 naming to docker.io/library/tp2-web:latest done
#11 unpacking to docker.io/library/tp2-web:latest 0.0s done
#11 DONE 0.4s

#5 [app internal] load metadata for docker.io/library/python:3.12
#5 DONE 1.3s

#13 [web] resolving provenance for metadata file
#13 DONE 0.0s

#14 [app internal] load .dockerignore
#14 transferring context: 2B done
#14 DONE 0.1s

#15 [app internal] load build context
#15 transferring context: 99B done
#15 DONE 0.1s

#16 [app 1/5] FROM docker.io/library/python:3.12@sha256:bd55e06128e2743d526738dd5cf5db924e49ead66d4fe2ad15f2aac8ad86aae9
#16 resolve docker.io/library/python:3.12@sha256:bd55e06128e2743d526738dd5cf5db924e49ead66d4fe2ad15f2aac8ad86aae9 0.1s done
#16 DONE 0.1s

#17 [app 2/5] WORKDIR /app
#17 CACHED

#18 [app 3/5] COPY requirements.txt .
#18 CACHED

#19 [app 4/5] RUN pip install -r requirements.txt
#19 CACHED

#20 [app 5/5] COPY . .
#20 CACHED

#21 [app] exporting to image
#21 exporting layers done
#21 exporting manifest sha256:d674f5b6b1a3cfe5b079f6b589aa14e99b5f1ff7c1692da69da55965129957c11 0.1s done
#21 exporting config sha256:f9b82e5b30f168da3d91ac559d41810e3a617ee267cc12bb08ea623f9f96e488 0.1s done
#21 exporting attestation manifest sha256:7eea9b4a91acc58439817f0b6e959e7dd5a4b533aa49c964fd841796adff2028 0.1s done
#21 exporting manifest list sha256:dff72d81238eaf8c2b79dc7ca228427765cc185fed96cd7887eb3477b0833154e
#21 exporting manifest list sha256:dff72d81238eaf8c2b79dc7ca228427765cc185fed96cd7887eb3477b0833154e 0.1s done
#21 naming to docker.io/library/tp2-app:latest done
#21 unpacking to docker.io/library/tp2-app:latest 0.0s done
#21 DONE 0.4s

#21 [app] resolving provenance for metadata file
#21 DONE 0.0s
[+] up 4/4
✔Image tp2-web Built 2.8s
✔Image tp2-app Built 2.8s
✔Container tp2-web-1 Started 1.4s
✔Container tp2-app-1 Started 1.3s
C:\Users\Gonza\Desktop\tps\DOCKER\tp2>

```

Ejercicio 8: Ejemplo práctico con múltiples servicios

Cree una aplicación en Flask que se conecte a una base de datos de su preferencia dentro de un docker compose.

docker-compose.yml:

```

1  version: "3.8"
2  services:
3      app:
4          build: ./app
5          networks:
6              - app_network
7      db:
8          image: mysql:5.7
9          environment:
10             MYSQL_ROOT_PASSWORD: example
11          networks:
12              - app_network
13  networks:
14      app_network:

```

archivo py:

```
app > app_code.py > ...
1 from flask import Flask
2 app = Flask(__name__)
3 @app.route('/')
4 def hello():
5     return "Hola, Docker"
6 if __name__ == "__main__":
7     app.run(host='0.0.0.0', port=5000)
```

ejecución:

```
C:\Users\Gonza\Desktop\tps\DOCKER\tp2-docker compose up
time="2026-05-30T11:49:41-03:00" level=warning msg="C:\Users\Gonza\Desktop\tps\DOCKER\tp2\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[*] up 17/17
✔Image mysql:5.7 Pulled 16.1s
✔Network tp2-app-network Created 0.1s
✔Container tp2-db-1 Created 17.6s
✔Container tp2-app-1 Created 17.6s
Attaching to app-1, db-1
db-1 | 2026-05-30 14:50:16+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
db-1 | 2026-05-30 14:50:16+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
db-1 | 2026-05-30 14:50:16+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 5.7.44-1.el7 started.
db-1 | 2026-05-30 14:50:17+00:00 [Note] [Entrypoint]: Initializing database files
db-1 | 2026-05-30T14:50:18.031303Z 0 [Warning] TIMESTAMP with implicit DEFAULT value is deprecated. Please use --explicit_defaults_for_timestamp server option (see documentation for more details).
db-1 | 2026-05-30T14:50:17.548245Z 0 [Warning] InnoDB: New log files created, LSN=45799
db-1 | 2026-05-30T14:50:17.897797Z 0 [Warning] InnoDB: Creating foreign key constraint system tables.
db-1 | 2026-05-30T14:50:17.976856Z 0 [Warning] No existing UUID has been found, so we assume that this is the first time that this server has been started. Generating a new UUID: dfe8e895-5c36-11f1-b0e3-fe08c
f50ed7b.
db-1 | 2026-05-30T14:50:17.993450Z 0 [Warning] Gtid table is not ready to be used. Table 'mysql.gtid_executed' cannot be opened.
db-1 | 2026-05-30 14:50:24+00:00 [Note] [Entrypoint]: A deprecated TLS version TLSv1 is enabled. Please use TLSv1.2 or higher.
db-1 | 2026-05-30T14:50:18.055647Z 0 [Warning] A deprecated TLS version TLSv1.1 is enabled. Please use TLSv1.2 or higher.
db-1 | 2026-05-30T14:50:18.055180Z 0 [Warning] CA certificate ca.pem is self signed.
db-1 | 2026-05-30T14:50:18.095134Z 1 [Warning] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
app-1 | * Serving Flask app 'pruebap2'
app-1 | * Debug mode: off
app-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
app-1 | * Running on all addresses (0.0.0.0)
app-1 | * Running on http://127.0.0.1:5000
app-1 | * Running on http://172.19.0.3:5000
app-1 | Press CTRL-C to quit
db-1 | 2026-05-30 14:50:24+00:00 [Note] [Entrypoint]: Database files initialized
db-1 | 2026-05-30 14:50:24+00:00 [Note] [Entrypoint]: Starting temporary server
db-1 | 2026-05-30 14:50:24+00:00 [Note] [Entrypoint]: Waiting for server startup
db-1 | 2026-05-30T14:50:24.820402Z 0 [Warning] TIMESTAMP with implicit DEFAULT value is deprecated. Please use --explicit_defaults_for_timestamp server option (see documentation for more details).
db-1 | 2026-05-30T14:50:24.828332Z 0 [Note] mysqld (mysqld 5.7.44) starting as process 125 ...
db-1 | 2026-05-30T14:50:24.828514Z 0 [Note] InnoDB: PUNCH HOLE support available
db-1 | 2026-05-30T14:50:24.828555Z 0 [Note] InnoDB: Mutexes and rw_locks use GCC atomic builtins
db-1 | 2026-05-30T14:50:24.828568Z 0 [Note] InnoDB: Uses event mutexes
db-1 | 2026-05-30T14:50:24.828563Z 0 [Note] InnoDB: GCC builtin __atomic_thread_fence() is used for memory barrier
db-1 | 2026-05-30T14:50:24.828565Z 0 [Note] InnoDB: Compressed tables use zlib 1.2.13
db-1 | 2026-05-30T14:50:24.828568Z 0 [Note] InnoDB: Using Linux native AIO
db-1 | 2026-05-30T14:50:24.828972Z 0 [Note] InnoDB: Number of pools: 1
db-1 | 2026-05-30T14:50:24.829181Z 0 [Note] InnoDB: Using CPU crc32 instructions
db-1 | 2026-05-30T14:50:24.832495Z 0 [Note] InnoDB: Initializing buffer pool, total size = 128M, instances = 1, chunk size = 128M
db-1 | 2026-05-30T14:50:24.845422Z 0 [Note] InnoDB: Completed initialization of buffer pool
db-1 | 2026-05-30T14:50:24.850736Z 0 [Note] InnoDB: If the mysqld execution user is authorized, page cleaner thread priority can be changed. See the man page of setpriority().
db-1 | 2026-05-30T14:50:24.864079Z 0 [Note] InnoDB: Highest supported file format is Barracuda.
db-1 | 2026-05-30T14:50:24.884601Z 0 [Note] InnoDB: Creating shared tablespace for temporary tables
db-1 | 2026-05-30T14:50:24.884798Z 0 [Note] InnoDB: Setting file './ibtmp1' size to 12 MB. Physically writing the file full; Please wait ...
db-1 | 2026-05-30T14:50:24.917345Z 0 [Note] InnoDB: File './ibtmp1' size is now 12 MB.
db-1 | 2026-05-30T14:50:24.918257Z 0 [Note] InnoDB: 96 redo rollback segment(s) found. 96 redo rollback segment(s) are active.
db-1 | 2026-05-30T14:50:24.918296Z 0 [Note] InnoDB: 32 non-redo rollback segment(s) are active.
db-1 | 2026-05-30T14:50:24.918571Z 0 [Note] InnoDB: 5.7.44 started; log sequence number 2768293
db-1 | 2026-05-30T14:50:24.919067Z 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buffer_pool
db-1 | 2026-05-30T14:50:24.920331Z 0 [Note] Plugin 'FEDERATED' is disabled.
db-1 | 2026-05-30T14:50:24.922566Z 0 [Note] InnoDB: Buffer pool(s) load completed at 20260530 14:50:24
db-1 | 2026-05-30T14:50:24.939977Z 0 [Note] Found ca.pem, server-cert.pem and server-key.pem in data directory. Trying to enable SSL support using them.
db-1 | 2026-05-30T14:50:24.931008Z 0 [Note] Skipping generation of SSL certificates as certificate files are present in data directory.
db-1 | 2026-05-30T14:50:24.931013Z 0 [Warning] A deprecated TLS version TLSv1 is enabled. Please use TLSv1.2 or higher.
db-1 | 2026-05-30T14:50:24.931016Z 0 [Warning] A deprecated TLS version TLSv1.1 is enabled. Please use TLSv1.2 or higher.
db-1 | 2026-05-30T14:50:24.931494Z 0 [Warning] CA certificate ca.pem is self signed.
db-1 | 2026-05-30T14:50:24.931563Z 0 [Note] Skipping generation of RSA key pair as key files are present in data directory.
db-1 | 2026-05-30T14:50:24.941371Z 0 [Warning] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
db-1 | 2026-05-30T14:50:24.948936Z 0 [Note] Event Scheduler: Loaded 0 events
db-1 | 2026-05-30T14:50:24.949436Z 0 [Note] mysqld: ready for connections.
db-1 | Version: '5.7.44' socket: '/var/run/mysqld/mysqld.sock' port: 0 MySQL Community Server (GPL)
db-1 | 2026-05-30 14:50:25+00:00 [Note] [Entrypoint]: Temporary server started.
db-1 | /var/lib/mysql/mysql.sock' -> /var/run/mysqld/mysqld.sock'
db-1 | 2026-05-30T14:50:25.625453Z 3 [Note] InnoDB: Stopping purge
db-1 | 2026-05-30T14:50:25.644622Z 3 [Note] InnoDB: Resuming purge
db-1 | 2026-05-30T14:50:25.653352Z 3 [Note] InnoDB: Stopping purge
db-1 | 2026-05-30T14:50:25.672520Z 3 [Note] InnoDB: Resuming purge
db-1 | 2026-05-30T14:50:25.681283Z 3 [Note] InnoDB: Stopping purge
db-1 | 2026-05-30T14:50:25.700583Z 3 [Note] InnoDB: Resuming purge
db-1 | 2026-05-30T14:50:25.709473Z 3 [Note] InnoDB: Stopping purge
db-1 | 2026-05-30T14:50:25.725978Z 3 [Note] InnoDB: Resuming purge
db-1 | Warning: Unable to load '/usr/share/zoneinfo/iso3166.tab' as time zone. Skipping it.
db-1 | Warning: Unable to load '/usr/share/zoneinfo/leapseconds' as time zone. Skipping it.
db-1 | Warning: Unable to load '/usr/share/zoneinfo/zone.tab' as time zone. Skipping it.
db-1 | Warning: Unable to load '/usr/share/zoneinfo/zone1970.tab' as time zone. Skipping it.
db-1 | 2026-05-30 14:50:28+00:00 [Note] [Entrypoint]: Stopping temporary server
db-1 | 2026-05-30T14:50:28.279808Z 0 [Note] Giving 0 client threads a chance to die gracefully
db-1 | 2026-05-30T14:50:28.279848Z 0 [Note] Shutting down slave threads
db-1 | 2026-05-30T14:50:28.279846Z 0 [Note] Forcefully disconnecting 0 remaining clients
db-1 | 2026-05-30T14:50:28.279851Z 0 [Note] Event Scheduler: Purging the queue. 0 events
db-1 | 2026-05-30T14:50:28.279913Z 0 [Note] Binlog end
db-1 | 2026-05-30T14:50:28.280331Z 0 [Note] Shutting down plugin 'ngram'
db-1 | 2026-05-30T14:50:28.280356Z 0 [Note] Shutting down plugin 'partition'
db-1 | 2026-05-30T14:50:28.280361Z 0 [Note] Shutting down plugin 'BLACKHOLE'
db-1 | 2026-05-30T14:50:28.280364Z 0 [Note] Shutting down plugin 'ARCHIVE'
db-1 | 2026-05-30T14:50:28.280366Z 0 [Note] Shutting down plugin 'PERFORMANCE_SCHEMA'
db-1 | 2026-05-30T14:50:28.280392Z 0 [Note] Shutting down plugin 'MRG_MYISAM'
db-1 | 2026-05-30T14:50:28.280395Z 0 [Note] Shutting down plugin 'MyISAM'
db-1 | 2026-05-30T14:50:28.280400Z 0 [Note] Shutting down plugin 'INNODB_SYS_VIRTUAL'
db-1 | 2026-05-30T14:50:28.280403Z 0 [Note] Shutting down plugin 'INNODB_SYS_DATAFILES'
db-1 | 2026-05-30T14:50:28.280404Z 0 [Note] Shutting down plugin 'INNODB_SYS_TABLESPACES'
db-1 | 2026-05-30T14:50:28.280406Z 0 [Note] Shutting down plugin 'INNODB_SYS_FOREIGN_COLS'
```

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db-1 2026-05-30T14:50:28.288409Z [Note] Shutting down plugin 'INNODB_SYS_FOREIGN'
db-1 2026-05-30T14:50:28.288412Z [Note] Shutting down plugin 'INNODB_SYS_FIELDS'
db-1 2026-05-30T14:50:28.288415Z [Note] Shutting down plugin 'INNODB_SYS_COLUMNS'
db-1 2026-05-30T14:50:28.288421Z [Note] Shutting down plugin 'INNODB_SYS_INDEXES'
db-1 2026-05-30T14:50:28.288423Z [Note] Shutting down plugin 'INNODB_SYS_TABLESTATS'
db-1 2026-05-30T14:50:28.288426Z [Note] Shutting down plugin 'INNODB_SYS_TABLES'
db-1 2026-05-30T14:50:28.288428Z [Note] Shutting down plugin 'INNODB_FT_INDEX_TABLE'
db-1 2026-05-30T14:50:28.288431Z [Note] Shutting down plugin 'INNODB_FT_INDEX_CACHE'
db-1 2026-05-30T14:50:28.288434Z [Note] Shutting down plugin 'INNODB_FT_CONFIG'
db-1 2026-05-30T14:50:28.288437Z [Note] Shutting down plugin 'INNODB_FT_BEING_DELETED'
db-1 2026-05-30T14:50:28.288440Z [Note] Shutting down plugin 'INNODB_FT_DELETED'
db-1 2026-05-30T14:50:28.288442Z [Note] Shutting down plugin 'INNODB_FT_DEFAULT_STOPWORD'
db-1 2026-05-30T14:50:28.288449Z [Note] Shutting down plugin 'INNODB_METRICS'
db-1 2026-05-30T14:50:28.288453Z [Note] Shutting down plugin 'INNODB_TEMP_TABLE_INFO'
db-1 2026-05-30T14:50:28.288456Z [Note] Shutting down plugin 'INNODB_BUFFER_POOL_STATS'
db-1 2026-05-30T14:50:28.288459Z [Note] Shutting down plugin 'INNODB_BUFFER_PAGE_L0'
db-1 2026-05-30T14:50:28.288461Z [Note] Shutting down plugin 'INNODB_BUFFER_PAGE'
db-1 2026-05-30T14:50:28.288464Z [Note] Shutting down plugin 'INNODB_CMP_PER_INDEX_RESET'
db-1 2026-05-30T14:50:28.288466Z [Note] Shutting down plugin 'INNODB_CMP_PER_INDEX'
db-1 2026-05-30T14:50:28.288469Z [Note] Shutting down plugin 'INNODB_CMPMEM_RESET'
db-1 2026-05-30T14:50:28.288472Z [Note] Shutting down plugin 'INNODB_CMPMEM'
db-1 2026-05-30T14:50:28.288475Z [Note] Shutting down plugin 'INNODB_CMP_RESET'
db-1 2026-05-30T14:50:28.288480Z [Note] Shutting down plugin 'INNODB_CMP'
db-1 2026-05-30T14:50:28.288484Z [Note] Shutting down plugin 'INNODB_LOCK_WAITS'
db-1 2026-05-30T14:50:28.288487Z [Note] Shutting down plugin 'INNODB_LOCKS'
db-1 2026-05-30T14:50:28.288490Z [Note] Shutting down plugin 'INNODB_TRX'
db-1 2026-05-30T14:50:28.288492Z [Note] Shutting down plugin 'InnoDB'
db-1 2026-05-30T14:50:28.288569Z [Note] InnoDB: FTS optimize thread exiting.
db-1 2026-05-30T14:50:28.288772Z [Note] InnoDB: Starting shutdown...
db-1 2026-05-30T14:50:28.310313Z [Note] InnoDB: Dumping buffer pool(s) to '/var/lib/mysql/ib_buffer_pool'
db-1 2026-05-30T14:50:28.310468Z [Note] InnoDB: Buffer pool(s) dump completed at 20260530 14:50:28
db-1 2026-05-30T14:50:30.126107Z [Note] InnoDB: Shutdown completed; log sequence number 12219419
db-1 2026-05-30T14:50:30.130836Z [Note] InnoDB: Removed temporary tablespace data file: "ibtmp1"
db-1 2026-05-30T14:50:30.130870Z [Note] InnoDB: Shutting down plugin 'HEURYS'
db-1 2026-05-30T14:50:30.130877Z [Note] InnoDB: Shutting down plugin 'CSV'
db-1 2026-05-30T14:50:30.130881Z [Note] InnoDB: Shutting down plugin 'sha256_password'
db-1 2026-05-30T14:50:30.130886Z [Note] InnoDB: Shutting down plugin 'mysql_native_password'
db-1 2026-05-30T14:50:30.130932Z [Note] InnoDB: Shutting down plugin 'binlog'
db-1 2026-05-30T14:50:30.133919Z [Note] mysqld: Shutdown complete
db-1
db-1 2026-05-30 14:50:30:00:00 [Note] [Entrypoint]: Temporary server stopped
db-1
db-1 2026-05-30 14:50:30:00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.
db-1
db-1 2026-05-30T14:50:30.095659Z [Warning] TIMESTAMP with implicit DEFAULT value is deprecated. Please use --explicit_defaults_for_timestamp server option (see documentation for more details).
db-1 2026-05-30T14:50:30.691576Z [Note] mysqld (mysqld 5.7.40) starting as process 1 ...
db-1 2026-05-30T14:50:30.505262Z [Note] InnoDB: PUNCH HOLE support available
db-1 2026-05-30T14:50:30.505399Z [Note] InnoDB: Mutexes and rw_locks use GCC atomic builtins
db-1 2026-05-30T14:50:30.505315Z [Note] InnoDB: Uses event mutexes
db-1 2026-05-30T14:50:30.505323Z [Note] InnoDB: GCC builtin __atomic_thread_fence() is used for memory barrier
db-1 2026-05-30T14:50:30.505326Z [Note] InnoDB: Compressed tables use zlib 1.2.13
db-1
db-1 2026-05-30T14:50:30.505328Z [Note] InnoDB: Using Linux native AIO
db-1 2026-05-30T14:50:30.505616Z [Note] InnoDB: Number of pools: 1
db-1 2026-05-30T14:50:30.505828Z [Note] InnoDB: Using CPU crc32 instructions
db-1 2026-05-30T14:50:30.506270Z [Note] InnoDB: Initializing buffer pool, total size = 128M, instances = 1, chunk size = 128M
db-1 2026-05-30T14:50:30.510655Z [Note] InnoDB: Completed initialization of buffer pool
db-1 2026-05-30T14:50:30.520283Z [Note] InnoDB: If the mysqld execution user is authorized, page cleaner thread priority can be changed. See the man page of setpriority().
db-1 2026-05-30T14:50:30.532422Z [Note] InnoDB: Highest supported file format is Barracuda.
db-1 2026-05-30T14:50:30.547397Z [Note] InnoDB: Creating shared tablespace for temporary tables
db-1 2026-05-30T14:50:30.547487Z [Note] InnoDB: Setting file './ibtmp1' size to 12 MB. Physically writing the file full; Please wait ...
db-1 2026-05-30T14:50:30.575383Z [Note] InnoDB: File './ibtmp1' size is now 12 MB.
db-1 2026-05-30T14:50:30.575362Z [Note] InnoDB: 96 redo rollback segment(s) found. 96 redo rollback segment(s) are active.
db-1 2026-05-30T14:50:30.575455Z [Note] InnoDB: 32 non-redo rollback segment(s) are active.
db-1 2026-05-30T14:50:30.575668Z [Note] InnoDB: 5.7.40 started; log sequence number 12219419
db-1 2026-05-30T14:50:30.575894Z [Note] InnoDB: Loading buffer pool(s) from '/var/lib/mysql/ib_buffer_pool'
db-1 2026-05-30T14:50:30.576466Z [Note] Plugin 'FEDERATED' is disabled.
db-1 2026-05-30T14:50:30.580352Z [Note] InnoDB: Buffer pool(s) load completed at 20260530 14:50:30
db-1 2026-05-30T14:50:30.583328Z [Note] Found ca.pem, server-cert.pem and server-key.pem in data directory. Trying to enable SSL support using them.
db-1 2026-05-30T14:50:30.583361Z [Note] Skipping generation of SSL certificates as certificate files are present in data directory.
db-1 2026-05-30T14:50:30.583367Z [Warning] A deprecated TLS version TLSv1 is enabled. Please use TLSv1.2 or higher.
db-1 2026-05-30T14:50:30.583370Z [Warning] A deprecated TLS version TLSv1.1 is enabled. Please use TLSv1.2 or higher.
db-1 2026-05-30T14:50:30.583387Z [Warning] CA certificate ca.pem is self signed.
db-1 2026-05-30T14:50:30.583915Z [Note] Skipping generation of RSA key pair as key files are present in data directory.
db-1 2026-05-30T14:50:30.584722Z [Note] Server hostnames (bind-address): '*'; port: 3306
db-1 2026-05-30T14:50:30.584699Z [Note] IPv6 is available.
db-1 2026-05-30T14:50:30.584697Z [Note] - '!' resolves to '!';;
db-1 2026-05-30T14:50:30.584532Z [Note] Server socket created on IP: '::'.
db-1 2026-05-30T14:50:30.593381Z [Warning] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
db-1 2026-05-30T14:50:30.610792Z [Note] Event Scheduler: Loaded 0 events
db-1 2026-05-30T14:50:30.610792Z [Note] mysqld: ready for connections.
db-1 Version: '5.7.40' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server (GPL)

```

en docker desktop:

The screenshot shows the Docker Desktop application window. At the top, there's a blue header with the Docker logo, 'docker desktop PERSONAL', a search bar, and several icons. Below the header, a light blue banner states 'A new version of Docker Desktop is available.' with an 'Update' button. The left sidebar contains a list of Docker features: Gordon (BETA), Containers, Images, Volumes, Kubernetes, Builds, Docker Hub, Docker Scout, Models, MCP Toolkit (BETA), and Extensions. The main area is titled 'Compose file viewer (BETA)' and shows the file path 'tp2:C:\Users\Gonza\Desktop\tps\DOCKER\tp2'. It displays a tree view on the left with 'tp2' expanded, showing 'docker-compose.yml' selected. The right pane shows the content of 'docker-compose.yml' with line numbers 1 through 14. The file content defines a service named 'app' with a build context of './app', a network of 'app_network', and a database service 'db' using the 'mysql:5.7' image with the environment variable 'MYSQL_ROOT_PASSWORD: example'. The 'networks' section defines 'app_network'. At the bottom of the window, a status bar shows 'Engine running' and system resources: 'RAM 1.19 GB CPU 0.33% Disk: 5.93 GB used (limit 1006.85 GB)'. There is also an 'Upgrade plan' button in the bottom left and an 'Update available' notification in the bottom right.